# Introduction to Computational Social Science

Ian Lundberg[1]

SICSS UCLA
18 Aug 2025

[1]Assistant Professor, Sociology, UCLA, ianlundberg@ucla.edu

# How computing looked **in the 1950s**



Source: NASA

# How computing looked **in the 1980s**



Source: Wikimedia

# How computing looks **today**




Source: Apple

# How computing looks **today**



Ian Lundberg

Can you write a poem about the Summer Institute in Computational Social Science at UCLA?

ChatGPT

Of course! Here's a poem about the Summer Institute in Computational Social Science at UCLA:

In sunny UCLA's embrace,
Where minds converge in shared space,
Computational marvels unfold,
At the Institute, stories untold.

Source: OpenAI

# What is computational social science?

# What is computational social science?

Many new forms of research in the digital age.

# What is computational social science?

Many new forms of research in the digital age.

Some common themes:

# What is computational social science?

Many new forms of research in the digital age.

Some common themes:

- computing as a source of new forms of data

# What is computational social science?

Many new forms of research in the digital age.

Some common themes:

- computing as a source of new forms of data
- computing as a subject of research

# What is computational social science?

Many new forms of research in the digital age.

Some common themes:

- computing as a source of new forms of data
- computing as a subject of research
- computing as a tool to answer old questions in new ways

# What is computational social science?

Many new forms of research in the digital age.

Some common themes:

- computing as a source of new forms of data
- computing as a subject of research
- computing as a tool to answer old questions in new ways
- and perhaps more

# What is computational social science?

Who is involved?

# What is computational social science?

Who is involved?

Applied statisticians

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers, political scientists

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers, political scientists, economists

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers, political scientists, economists, policymakers

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers, political scientists, economists, policymakers, corporate leaders

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers, political scientists, economists, policymakers, corporate leaders, community leaders

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers, political scientists, economists, policymakers, corporate leaders, community leaders, ethics experts

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers, political scientists, economists, policymakers, corporate leaders, community leaders, ethics experts, and more

# What is computational social science?

Who is involved?

Applied statisticians, computer scientists, survey methodologists, psychologists, demographers, sociologists, ethnographers, political scientists, economists, policymakers, corporate leaders, community leaders, ethics experts, and more

Computational social science benefits from **everyone's ideas**

# Summer Institute in Computational Social Science

- Began in 2017
- Expanded to sites around the world

# Summer Institute in Computational Social Science

**What is SICSS all about?**

# Summer Institute in Computational Social Science

**What is SICSS all about?**

The purpose of the Summer Institutes is to bring together graduate students, postdoctoral researchers, and beginning faculty interested in computational social science.

(from sicss.io)

SICSS is all about **you**

# SICSS is all about **you**

What excites you about computational social science?

- perhaps new forms of data
- perhaps new ways of analyzing traditional data
- perhaps something else

How might it connect to your research?